The OSSEC system comprises the following three main components.

- **Main application:** This is a prime requirement for installations; OSSEC is supported by Linux, Windows, Solaris and Mac environments.
- **Windows agent:** This is only required when OSSEC is to be installed on Windows based computers/clients as well as servers.
- **Web interface:** Web based GUI application for defining rules and network monitoring.

Pros:
- Multi-platform IDS system providing real-time and configurable alerts.
- Centralised management, with agent and agentless monitoring.
- Can be used both in serverless and server-agent mode.

Cons:
- Upgrade process overwrites existing rules with out-of-the-box rules.
- Pre-sharing keys can be troublesome.
- Windows OS is only supported in server-agent mode.

Latest version: 2.8.3
Official website: *http://ossec.github.io/*

Open Source Tripwire

Open Source Tripwire is a host based intrusion detection system focusing on detecting changes in file system objects. On the first initialisation, Tripwire scans the file system as instructed by the systems administrator and stores the information of each file in a database. When files are changed and on future scans, the results are compared with the stored values and changes are reported to users.

Tripwire makes use of cryptographic hashes to detect changes in files. In addition to scanning file changes, it is used for integrity assurance, change management and policy compliance.

Pros:
- Excellent for small, decentralised Linux systems.
- Good integration with Linux.

Cons:
- Only runs on Linux.
- Requires the user to be a Linux expert.
- Advanced features are not available in open source versions.
- No real-time alerts.

Latest version: 2.4.3.1
Official website: *https://github.com/Tripwire/tripwire-open-source*

AIDE

AIDE (Advanced Intrusion Detection Environment) was developed by Rami Lehti and Pablo Virolainen. It is regarded as one of the most powerful tools for monitoring changes to UNIX or Linux systems. AIDE creates a database via regular expression rules that it finds from the *config* files. On initialising the database, it is used to verify the integrity of files.

Some of the most powerful features of AIDE are as follows:
- Supports all kinds of message digest algorithms like MD5, SHA1, RMD160, TIGER, SHA256 and SHA512.
- Supports POSIX ACL, SELinux, XAttra and Extended File System.
- Powerful regular expression support to include or exclude files and directories for monitoring.
- Supports various operating system platforms like Linux, Solaris, Mac OS X, UNIX, BSD, HP-UX, etc.

Pros:
- Real-time detection and elimination of the attacker to restore file or directory properties.
- Anomaly detection to reduce the false rate of file system monitors.
- Supports a wide range of encryption algorithms.

Cons:
- No GUI interface.
- Requires careful configuration for effective detection and prevention.
- Doesn't deal properly with long file names for smooth detection.

Latest version: 0.16
Official website: *http://aide.sourceforge.net/*

END

References

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